ExaThon India 2026 · Track 02: Customers, Channels & Demand

ShelfSense

Catch stock-outs before they happen.

Stock-out early warning for quick-commerce dark stores, computed entirely in SQL inside Exasol Personal.

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shelfsense-exathon.vercel.app · github.com/andringodson/shelfsense

An empty shelf is a lost customer

- The customer doesn't wait. They open another app.
- Store managers find out from complaints or a weekly report, hours too late.
- Sales data can't show it: an empty shelf at 7 PM looks exactly like a quiet hour.

5.7%

of all demand goes unserved because the shelf was empty

₹15 lakh

revenue lost in 14 days across 25 stores (simulated network)

Stock-outs hide, then breed more stock-outs

1

Shelf runs empty

Sales record 0, not the 12 customers who wanted it.

2

Forecast learns "low demand"

Models trained on sales copy the stock-out zeros.

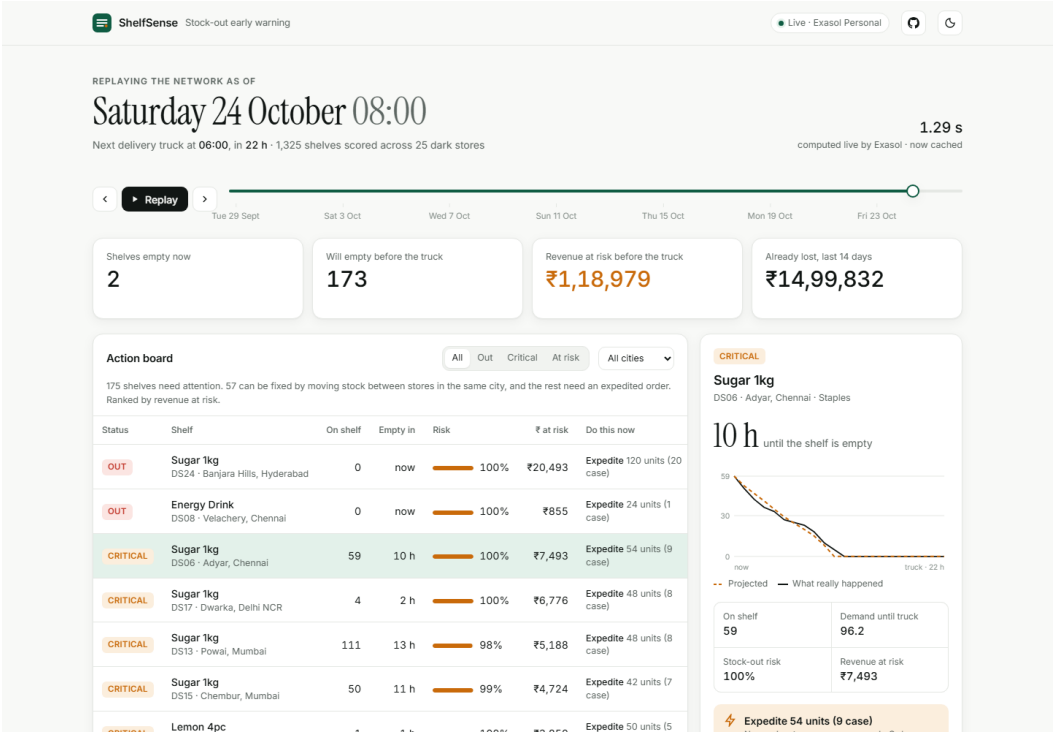
3

Next order is smaller

So the shelf empties earlier, and the loop tightens.

ShelfSense breaks the loop: it learns demand only from hours when the shelf was NOT empty.

Will this shelf empty before the truck?



1 **Action board**

Risk of running empty before the 06:00 truck, revenue at stake, and the fix: transfer or expedite.

2 **Lost sales**

Demand hidden behind every empty-shelf hour, valued in ₹.

3 **Backtest replay**

Rewind to any hour, then score the calls against what really happened.

All the intelligence lives in Exasol as SQL views

V_SHAPE · V_RATE

Censoring-aware demand

V_FORECAST

48-hour forecast

V_RISK

Poisson risk, hours to empty

V_ACTIONS

Same-city rebalancing

V_BACKTEST

Scored against the future

1.3 M

hourly rows bulk-loaded in 2.5 s

1.2 s

to score every shelf at any hour: 3 queries, on a laptop

20 ms

to replay any hour once cached: the timeline scrubs live

The calls hold up against what really happened

72%

of alerts were right (precision)

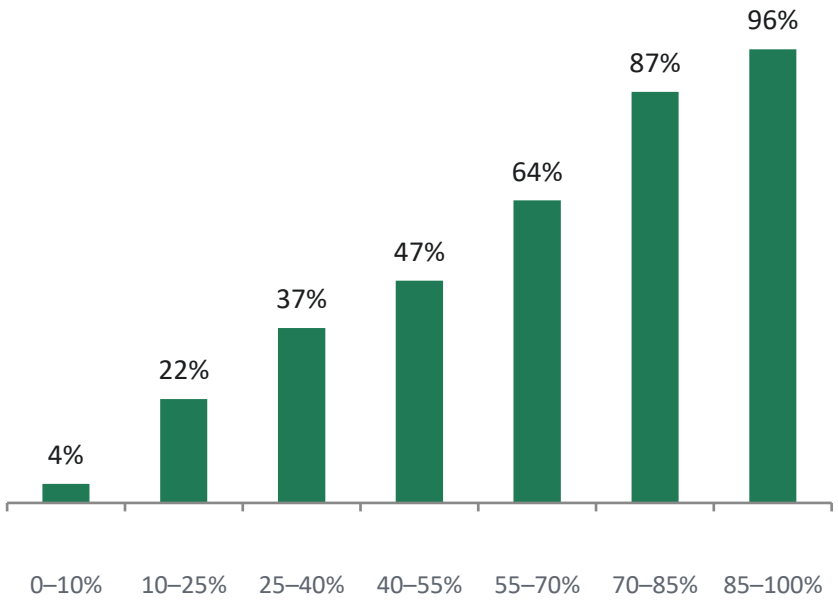
57%

of real stock-outs caught in advance

96%

of shelves rated 85–100% risk really ran out. At 0–10% risk, only 4% did.

Predicted risk vs actual stock-out rate



From alert to action in one screen

Saturday, 08:00 · 25 stores

173 shelves

will empty before tomorrow's truck

₹1.19 lakh

at risk today, before anyone notices

What the ops team gets

- 57 shortfalls fixed by a same-city transfer: "Transfer 23 from DS07 (T Nagar)"
- The rest get a case-pack-rounded expedite order
- Ranked by ₹ at risk, so the first call is the most valuable one
- Planners see true demand, so the next order stops shrinking

Real-world next steps

1

Plug in real feeds

POS and WMS exports load with the same IMPORT path. The schema is 4 tables.

2

Smarter fixes

Dark-store distance, rider capacity, and shelf life for perishables.

3

Push, not pull

Alerts to store managers on WhatsApp or Slack the moment risk crosses 70%.

Live: shelfsense-exathon.vercel.app
Code: github.com/andringodson/shelfsense (MIT)

Andrin Godson: data model, Exasol SQL, dashboard
Karthikeyan: testing, reproducibility, demo review